Guinea-Bissau—External Debt Sustainability Analysis[23]

Guinea-Bissau is a country in severe debt distress. However, assuming that HIPC completion point is achieved by the end of 2009, the risk of debt distress is expected to fall to more moderate levels. Specifically, external debt would be reduced to levels that are sustainable under the baseline scenario, but the country would still be vulnerable to exchange rate depreciation, a reduction in the growth rate or a reduction in the grant content of new loans. The total public debt to GDP ratio remains well above threshold levels even in the baseline scenario, reflecting the weight of capital contributions to WAEMU.

Introduction

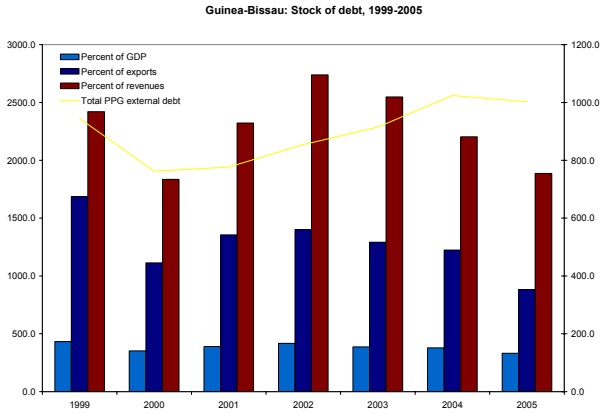
75. After reaching the decision point of the HIPC initiative in 2000, Guinea-Bissau failed to maintain macroeconomic stability and progress on the implementation of its poverty reduction strategy. The PRGF-supported program went off track in 2001, reflecting fiscal policy slippages associated with military spending, an increase in the wage bill and political interference in public finance management. The preparation of the full PRSP was delayed as a result of political instability and capacity constraints. In March 2005, the government engaged in an SMP, covering April–December 2005, focused on stabilizing the fiscal situation, including avoiding new domestic arrears, promoting economic growth by rebuilding infrastructure and removing excessive regulation, and improving governance. Progress under the SMP was mixed, with the wage bill and domestic arrears exceeding indicative targets.

76. **In 2006, the government agreed to a new timeline to reengage in programs supported by the IMF and the World Bank, aiming to reach the completion point under the HIPC Initiative by end-2009.** A new SMP to cover 2006 was discussed with the authorities in March, to help the authorities establish a track record of policy implementation and address governance and public expenditure management issues. At end-2006, the government is also expected to finalize its full PRSP in order to start implementing it in 2007. Assuming the emergence of a concerted international assistance effort, Guinea-Bissau would receive support from the IMF under the Emergency Post- Conflict Country Assistance (EPCA) facility. The EPCA should pave the way for a new PRGF supported program by end-2008. The World Bank would complete a public expenditure review update and a fiduciary assessment by mid-2006 and support Guinea-Bissau in FY2007–08 through an Interim Strategy Note, focusing on infrastructure rehabilitation, reforms in the agricultural sectors and in economic management. Guinea-Bissau is expected to reach the completion point in the fourth quarter of 2009.

[23] Prepared under the joint debt sustainability framework for low income countries of the IMF and the World Bank. This analysis focuses on public external and domestic debt. Data on private external debt are not available and are therefore excluded from the analysis.

Public debt situation as of end-December 2005

77. **At end-2005, Guinea-Bissau's stock of external public and publicly guaranteed (PPG) external debt amounted to $1,002 million, of which $327 million in arrears.**[24] Multilateral debt amounts to 49 percent of total PPG debt (14 percent is owed to AfDB/AfDF, 29 percent is owed to IDA, 1.2 percent is owed to IMF) and bilateral and commercial amounts to 51 percent (Table 1). The NPV of debt amounts to $745.8 million or 247.3 percent of GDP and 656.9 percent of exports. Despite the concessional nature of most of the PPG external debt, the debt burden indicators far exceed the policy dependent debt thresholds for Guinea-Bissau.[25]



78. **The nominal external debt to GDP ratio has decreased slightly over the last five years from 353 percent at end-2000 to 332 percent at end-2005.** This positive development, however, is not due to a significant increase in the current account balance or in FDI, either of which would reduce the financing needs of the country. Instead, this debt to GDP reduction was due mainly to a strong appreciation of the CFA franc (pegged to the euro) with respect to the dollar, which translated into high growth rates of nominal GDP in U.S. dollar terms, despite low real GDP growth rates.

79. **Nominal public domestic debt amounted to 48 percent of GDP at end-2005.** Nearly 90 percent of this debt corresponds to a required capital contribution to join WAEMU for a total of CFAF 70 billion to be paid over 25 years starting in 2005. The remaining domestic debt is composed of obligations to BCEAO linked to statutory advances (no longer allowed) and to the failure of two banks after the civil war. Domestic arrears are not included in this debt estimate as an audit will be necessary for a fairly accurate estimation of these obligations to be possible.

80. **Since the PRGF went off track, Guinea-Bissau has not benefited from most of the debt relief committed at decision point.** At the decision point in 2000, creditors representing 80 percent of Guinea-Bissau external debt pledged to provide HIPC relief amounting to $790 million ($417 million in NPV terms). Since 2001, Guinea-Bissau has had to service a large share of external debt in full. The IMF suspended the provision of interim

[24] Bilateral debt data are provided by the authorities, while multilateral debt data are reconciled with creditors.

[25] According to the World Bank Country Policy and Institutions Assessment (CPIA), Guinea Bissau has a poor quality of policies and institutions. The CPIA rating for 2005 is 2.7, on a scale from 1 to 6, below the operational cutoff of 3.25 for poor performers.

relief and the Paris Club declared null and void any debt rescheduling agreements beyond end-2001. Many agreements signed with other multilateral and bilateral creditors have not been implemented because the country failed to remain current on debt service obligations.[26] IDA and the African Development Bank (AfDB) Group continued providing interim relief,[27] but both multilateral banks will reach the statutory ceiling for the delivery of interim relief at end-2007 and end-2006 respectively.[28] Only China and Cuba cancelled all outstanding claims.

81. **After the decision point, Guinea-Bissau could not service its external debt and accumulated arrears to most of its external creditors.** Since 2001, the country has not repaid any creditor that did not provide interim relief, with the exception of the IMF. The stock of arrears has increased from $139 million before decision point to $327 million at end-2005. In the following analysis, arrears are assumed to be cleared through concessional rescheduling before reaching the completion point. However, the HIPC process would be delayed if additional arrears arise to multilateral creditors or if Guinea- Bissau could not agree on a repayment schedule of post-cutoff-date arrears with the Paris Club.

Macroframework

8. Projections for the evolution of external and total public debt and debt service are carried out in a baseline scenario and several alternative scenarios: the baseline scenario assumes that the country stays on track with IMF programs and successfully reengages with the donor community; a historic scenario assumes key variables at their 10-year historical average; and different bound test scenarios, simulating temporary shocks to growth, exports, fiscal revenues, the exchange rate, prices and financial flows.

9. The macroeconomic assumptions underlying the baseline projection scenario up to 2009 are described in detail in Box 1. The baseline assumes that an international concerted assistance effort is in place by end-2006 to help finance urgently needed structural reforms in the public sector. Post-conflict assistance is assumed to be provided by the Fund by end-2006, followed, after two years, by a PRGF. Growth, at 5 percent by 2008, will be sustained by reforms in the electricity sector and investment in infrastructure which would loosen the bottleneck for private investment and sustained exports growth. The current account deficit is

[26] See IMF Country Report 05/93, November 2004, Box 10, for a comprehensive list of debt rescheduling agreements reached by Guinea-Bissau with the creditors as part of the provision of HIPC relief.

[27] Interim relief from IDA amounted to 100 percent of debt service falling due on debt disbursed before end-1999 until October 2003, the expected date of the completion point. Interim relief was decreased to 90 percent of debt service thereafter. The AfDB extended 100 percent debt service relief to end-2006.

[28] IDA could at its discretion extend interim relief to 50 percent of total HIPC relief based on satisfactory progress toward the completion point triggers and performance under an IMF-supported program. The AfDB could raise the interim relief ceiling from 40 percent to 50 percent.

projected initially to deteriorate, but it would be contained by strong export growth and fiscal consolidation in the medium and long term. The primary fiscal balance is initially expected to deteriorate as more resources are assigned to public investment and poverty reduction. In the medium term, however, stronger revenue performance and civil and security service reforms would contain the public sector wage bill and improve the fiscal balance. Public external borrowing at concessional terms would be contained over time, while domestic debt and arrears would be progressively paid down.

10. HIPC Completion Point is assumed by end-2009. Total HIPC debt relief, as established at decision point in 2000, amounts to US$ 413 million in NPV terms and would reduce debt service payments by about 7 percent of GDP per year.

Box 1: Macroeconomic Assumptions Under Baseline Scenario

- GDP growth rate increases from 3.5 percent in 2005 to 5 percent by 2008, and is sustained over the following years.
- GDP deflator is assumed to grow at a rate slightly below CPI inflation, due to expected worsening in the terms of trade. After 2009, both GDP deflator and CPI are assumed to increase at a rate of 2 percent.
- The noninterest current account is assumed to deteriorate over the medium term, reflecting both an increase in foreign-financed investment projects and lower official transfers in the form of budget support. Over the long run, the noninterest current account balance as a proportion of GDP converges back to its 2005 level. Exports are assumed to grow at around 8 percent per year until 2010, driven by steady growth in cashew nut production at least in the medium-term; export growth rate is assumed to decrease to about 6 percent thereafter.
- Foreign direct investment as a ratio of GDP ratio is assumed to remain constant in the medium and long term at the level observed in 2005 (1.3 percent).
- New borrowing will be contracted at highly concessional terms, with the grant element in new disbursements assumed to remain slightly above the 50 percent that was agreed on under the 2006 SMP.
- The primary fiscal deficit is assumed to decrease from 4.5 percent of GDP in 2006 to 1.7 percent by 2010, maintaining this level from then on.
- Total external loan financing is assumed to be 5 percent of GDP per year on average over the next 20 years.

Debt sustainability

11. The sustainability of the current external and total public debt is judged based on various indicative debt and debt service thresholds established for countries with poorly performing economic policies and institutions, and shown in Table 1.

Table 1: Summary of Baseline Debt Sustainability Indicators [1/]

	Indicative Threshold[2/]	2005	2016	2026	Average 2006-24
NPV of debt-to-GDP	30	247.3	35.9	*25.0*	68.8
NPV of debt-to-exports	100	656.9	*86.9*	*62.0*	168.4
NPV of debt-to-revenue	200	1482.3	*171.8*	*130.8*	456.3
Debt service-to-exports	15	43.6	*4.5*	*3.6*	*8.5*
Debt service-to-revenue	25	94	*11.9*	*10.6*	*19.8*

Note: Figures in italics are under the indicative thresholds.
[1/] All debt indicators assume the full delivery of HIPC relief at end-2009 and refer to the NPV of public and publicly guaranteed external debt.
[2/] Threshold over which countries with similar evaluations of policies and institutions would have at least a 25 percent chance of having a prolonged incident of debt distress in the coming year. Guinea-Bissau lies within the bottom quintile of countries ranked by the World Bank's Country Policy and Institutional Assessment Index (CPIA).

12. The simulations presented in Tables 2a, 2b, 3a, 3b and Figures 1 and 2, point to a country currently in severe debt distress, with arrears accruing to multilateral creditors, Paris Club creditors and other official and commercial creditors. Such arrears are projected to be settled by 2009 only if the HIPC process remains on track. Assuming HIPC relief, indicators based on debt and debt service ratios in simulated scenarios point to mixed signals of the risk of debt distress. The NPV of external debt-to-GDP ratio remains above the 30 percent threshold during most of the simulation period in the baseline scenario as well as under the stress test scenarios. In the case of total public debt, the debt to GDP ratio remains well above the threshold levels for the whole period and the ratio to government revenues surpasses the threshold for a significant period, reflecting the weight of capital contribution commitments to WAEMU on top of the external debt burden. On the other hand, the debt service ratios to exports and government revenues quickly fall below threshold levels in the baseline scenario immediately after the HIPC completion point is reached.

13. Alternative scenarios and stress tests indicate that significant exchange rate depreciation, reductions in GDP growth, and a lower grant element of new loans would substantially increase the risk of debt distress. The most severe scenario is that created by the effects of large (30 percent) exchange rate devaluation, which would cause the NPV of external debt to GDP ratio to significantly exceed the 30 percent threshold during the whole projection period.

14. These simulations do not include the additional debt relief that could be obtained from the Multilateral Debt Relief Initiative (MDRI), which would amount, at completion point, to US$77.3 million in net present value, or 21 percent of GDP. In this case, the external debt service to exports ratio would be reduced by an additional 3 percentage points.

Staffs appraisal

15. Since Guinea-Bissau is running arrears, it is automatically rated as in debt distress. Guinea-Bissau's risk of debt distress would be reduced substantially if the HIPC completion point were achieved by end-2009, as hoped for under current arrangements. Continued political stability, sustained adjustment, and reform, and a concerted international assistance effort are key elements for the country to achieve completion point and the associated debt reduction to sustainable levels. Even within this scenario, the country's capacity to sustain reasonable debt burden ratios may be compromised by significant shocks to the nominal exchange rate, the growth rate, or the concessionality of new borrowing.

Table 2a. Guinea-Bissau: External Debt Sustainability Framework, Baseline Scenario, 2006-2026 1/
(In percent of GDP, unless otherwise indicated)

		Actual					Historical	Standard	Projections									
	#	2001	2002	2003	2004	2005	Average 6/	Deviation 6/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
External debt (nominal) 1/	**...**	**390.0**	**417.9**	**387.2**	**378.9**	**332.2**			**317.4**	**296.6**	**276.6**	**118.6**	**107.0**	**96.4**		**54.3**	**13.1**	
o/w public and publicly guaranteed (PPG)	...	390.0	417.9	387.2	378.9	332.2			317.4	296.6	276.6	118.6	107.0	96.4		54.3	13.1	
Change in external debt	...	37.3	27.9	-30.7	-8.2	-46.7			-14.8	-20.8	-20.0	-158.0	-11.6	-10.6		-6.9	-2.5	
Identified net debt-creating flows	...	51.2	-0.2	-55.3	-55.1	-36.0			-10.3	-9.4	-10.3	-10.2	-4.5	-4.0		-2.3	-0.7	
Non-interest current account deficit	**...**	**14.1**	**6.4**	**-2.5**	**-7.3**	**2.9**	**2.3**	**7.4**	**3.7**	**5.9**	**4.9**	**4.4**	**4.0**	**3.9**		**3.3**	**3.3**	3.4
Deficit in balance of goods and services	...	34.3	21.4	14.1	16.2	17.4			19.6	18.4	17.5	16.9	15.2	15.2		15.2	15.1	
Exports	...	28.8	29.8	30.0	30.9	37.6			39.1	40.0	40.8	41.5	42.3	42.2		41.4	40.4	
Imports	...	63.1	51.2	44.1	47.1	55.1			58.7	58.4	58.3	58.5	57.5	57.4		56.5	55.5	
Net current transfers (negative = inflow)	...	-20.1	-15.0	-16.6	-23.5	-14.3	-18.2	3.5	-15.8	-12.5	-12.5	-12.4	-12.4	-12.4		-12.4	-12.4	-12.4
Other current account flows (negative = net inflow)	...	0.0	0.0	0.0	0.0	-0.2			0.0	0.0	-0.1	-0.1	1.2	1.1		0.5	0.6	
Net FDI (negative = inflow)	**...**	**-0.8**	**-1.1**	**-1.5**	**-3.5**	**-3.7**	**-1.8**	**1.4**	**-3.7**	**-3.7**	**-3.7**	**-3.7**	**-3.7**	**-3.7**		**-3.7**	**-3.7**	-3.7
Endogenous debt dynamics 2/	**...**	**37.9**	**-5.5**	**-51.4**	**-44.3**	**-35.2**			**-10.4**	**-11.7**	**-11.6**	**-11.0**	**-4.8**	**-4.3**		**-1.9**	**-0.4**	
Contribution from nominal interest rate	...	8.0	4.3	5.3	4.2	4.2			3.0	2.6	2.4	2.1	0.7	0.7		0.9	0.4	
Contribution from real GDP growth	...	-0.7	27.2	2.3	-7.5	-11.8			-13.4	-14.3	-13.9	-13.1	-5.6	-5.0		-2.9	-0.7	
Contribution from price and exchange rate changes	...	30.7	-37.0	-58.9	-41.0	-27.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**...**	**-14.0**	**28.1**	**24.6**	**46.8**	**-10.7**			**-4.5**	**-11.4**	**-9.7**	**-147.8**	**-7.1**	**-6.6**		**-4.6**	**-1.7**	
o/w exceptional financing	...	-21.6	-17.5	-16.6	-16.8	-14.3			-9.9	-7.8	-7.4	-148.2	-3.4	-3.1		-2.2	-0.8	
NPV of external debt 4/	...	...	...	...	...	247.3			239.2	229.8	223.6	145.1	69.4	40.4		35.9	25.0	
In percent of exports	...	...	...	...	...	656.9			611.2	574.3	548.5	349.2	164.0	95.7		86.9	62.0	
NPV of PPG external debt	**...**	**...**	**...**	**...**	**...**	**247.3**			**239.2**	**229.8**	**223.6**	**145.1**	**69.4**	**40.4**		**35.9**	**25.0**	
In percent of exports	...	...	...	...	...	656.9			611.2	574.3	548.5	349.2	164.0	95.7		86.9	62.0	
Debt service-to-exports ratio (in percent)	...	69.1	60.9	64.0	56.6	44.0			31.9	28.6	25.1	21.6	4.5	4.4		4.5	3.6	
PPG debt service-to-exports ratio (in percent)	...	69.1	60.9	64.0	56.6	44.0			31.9	28.6	25.1	21.6	4.5	4.4		4.5	3.6	
Total gross financing need (billions of U.S. dollars)	...	66.1	48.0	35.9	18.2	47.6			56.4	45.3	40.6	36.4	8.8	8.9		8.9	13.2	
Non-interest current account deficit that stabilizes debt ratio	...	-23.1	-21.5	28.3	1.0	49.6			18.5	26.7	24.9	162.4	15.6	14.5		10.2	5.8	
Key macroeconomic assumptions																		
Real GDP growth (in percent)	...	0.2	-7.1	-0.6	2.2	3.5	-0.4	4.1	4.2	4.8	5.0	5.0	5.0	5.0	4.8	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	...	-8.0	10.5	16.4	11.8	7.9	7.7	9.3	-0.4	1.1	1.2	0.9	1.2	1.5	0.9	2.0	2.0	2.0
Effective interest rate (percent) 5/	...	2.1	1.1	1.5	1.3	1.2	1.4	0.3	0.9	0.9	0.8	0.8	0.7	0.7	0.8	1.6	2.5	1.6
Growth of exports of G&S (US dollar terms, in percent)	...	-16.2	6.2	16.2	18.1	35.8	12.0	19.0	7.8	8.3	8.2	8.0	8.3	6.2	7.8	6.7	7.1	6.8
Growth of imports of G&S (US dollar terms, in percent)	...	13.0	-16.7	-0.4	22.2	30.4	9.7	18.7	10.5	5.5	6.0	6.3	4.6	6.2	6.5	6.8	7.1	6.9
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	...	...	56.9	56.9	43.0	51.0	51.1	56.9	52.6	56.9	56.9	56.9
Memorandym item:																		
Nominal GDP (millions of US dollars)	...	199.2	204.3	236.4	270.2	301.6			312.9	331.5	352.1	373.2	396.8	422.8		595.7	1182.9	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2b. Guinea-Bissau: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(In percent)

				Projections				
	2006	2007	2008	2009	2010	2011	**2016**	2026
NPV of debt-to-GDP ratio								
Baseline	239	230	224	145	69	40	**36**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	239	226	217	136	63	35	**32**	24
A2. New public sector loans on less favorable terms in 2007-26 2/	239	231	227	150	75	47	**47**	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	239	252	270	175	84	49	**43**	30
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	239	232	231	152	76	47	**42**	27
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	239	236	236	153	73	43	**38**	26
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	239	230	225	146	70	41	**37**	25
B5. Combination of B1-B4 using one-half standard deviation shocks	239	242	249	162	78	45	**40**	28
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	239	327	318	206	99	57	**51**	36
NPV of debt-to-exports ratio								
Baseline	611	574	548	349	164	96	**87**	62
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	611	565	533	328	148	83	**76**	61
A2. New public sector loans on less favorable terms in 2007-26 2/	611	578	556	360	178	112	**114**	94
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	611	574	548	349	164	96	**87**	62
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	611	677	768	497	245	152	**137**	92
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	611	574	548	349	164	96	**87**	62
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	611	575	551	351	166	98	**89**	63
B5. Combination of B1-B4 using one-half standard deviation shocks	611	606	613	391	184	108	**98**	70
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	611	574	548	349	164	96	**87**	62
Debt service-to-exports ratio								
Baseline	32	29	25	22	4	4	**4**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	32	28	25	21	4	4	**4**	3
A2. New public sector loans on less favorable terms in 2007-26 2/	32	29	25	22	5	6	**5**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	32	29	25	22	4	4	**4**	4
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	32	33	34	30	6	6	**7**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	32	29	25	22	4	4	**4**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	32	29	25	22	4	4	**5**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	32	30	28	24	5	5	**5**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	32	29	25	22	4	4	**4**	4
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	55	55	55	55	55	55	**55**	55

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3a.Country: Public Sector Debt Sustainability Framework, Baseline Scenario, 2003-2026
(In percent of GDP, unless otherwise indicated)

	Actual							Estimate	Projections								
	2001	2002	2003	2004	2005	Historical Average 5/	Standard Deviation 5/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2026	2012-26 Average
Public sector debt 1/	445.8	475.1	444.9	433.0	379.8			360.1	336.1	313.2	152.6	139.0	126.5		75.6	23.8	
o/w foreign-currency denominated	390.0	417.9	387.2	378.9	332.2			317.4	296.6	276.6	118.6	107.0	96.4		54.3	13.1	
Change in public sector debt	40.0	29.3	-30.2	-11.9	-53.2			-19.6	-24.0	-22.9	-160.6	-13.6	-12.6		-8.4	-3.2	
Identified debt-creating flows	31.7	-60.0	-59.9	-46.0	7.6			-35.6	-21.0	-20.4	-160.7	-8.8	-8.2		-5.1	-0.7	
Primary deficit	3.4	6.5	10.0	8.9	7.8	7.0	2.4	4.5	4.2	4.2	3.4	1.7	1.7	3.3	1.7	1.6	1.7
Revenue and grants	31.3	21.4	22.9	34.4	26.0			32.1	28.6	28.9	29.0	29.0	28.9		28.2	27.0	
of which : grants	14.5	6.1	7.7	17.2	8.4			11.4	8.7	8.7	8.6	8.6	8.6		8.4	8.0	
Primary (noninterest) expenditure	34.6	27.9	32.9	43.3	33.9			36.6	32.8	33.1	32.4	30.8	30.6		29.9	28.6	
Automatic debt dynamics	50.4	-47.4	-52.0	-38.3	12.9			-29.9	-17.4	-17.1	-15.5	-7.1	-6.8		-4.5	-1.5	
Contribution from interest rate/growth differential	1.6	29.3	1.7	-14.2	-23.9			-21.8	-18.2	-18.0	-17.7	-7.8	-7.2		-4.3	-1.4	
of which : contribution from average real interest rate	2.4	-5.0	-1.3	-4.6	-9.4			-6.5	-3.8	-3.6	-3.4	-0.7	-0.8		-0.7	-0.2	
of which : contribution from real GDP growth	-0.8	34.3	3.0	-9.6	-14.5			-15.2	-14.4	-14.4	-14.3	-7.0	-6.4		-3.6	-1.2	
Contribution from real exchange rate depreciation	48.8	-76.7	-53.7	-24.1	36.8			-8.1	0.9	1.0	2.1	0.7	0.3		...	...	
Other identified debt-creating flows	-22.1	-19.1	-17.9	-16.6	-13.1			-10.3	-7.9	-7.5	-148.5	-3.4	-3.1		-2.2	-0.8	
Privatization receipts (negative)	-1.3	-0.3	0.0	0.0	-0.1			-0.3	-0.1	-0.1	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-20.8	-18.8	-17.9	-16.6	-13.0			-10.0	-7.8	-7.4	-148.5	-3.4	-3.1		-2.2	-0.8	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.3	89.3	29.8	34.1	-60.9			16.0	-3.0	-2.6	0.0	-4.8	-4.3		-3.3	-2.6	
NPV of public sector debt	...	...	...	...	308.6			281.9	269.4	260.2	179.0	101.4	70.4		57.3	35.8	
o/w foreign-currency denominated	...	...	...	...	261.1			239.2	229.8	223.6	145.1	69.4	40.4		35.9	25.0	
o/w external	...	...	...	...	261.1			239.2	229.8	223.6	145.1	69.4	40.4		35.9	25.0	
NPV of contingent liabilities (not included in public sector debt)	...	...	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	23.7	25.2	29.5	28.3	24.2			22.8	15.8	14.6	12.4	4.8	4.5		3.6	3.0	
NPV of public sector debt-to-revenue ratio (in percent) 3/	...	...	...	...	1185.7			877.8	943.3	900.7	618.3	349.5	243.9		203.1	132.4	
o/w external	...	...	...	...	1003.1			744.8	804.8	773.9	501.0	239.2	139.8		127.5	92.7	
Debt service-to-revenue ratio (in percent) 3/ 4/	64.9	87.4	85.0	56.4	63.0			39.0	40.6	35.8	31.0	10.5	9.8		6.9	5.3	
Primary deficit that stabilizes the debt-to-GDP ratio			40.2	20.8	61.1			24.2	28.2	27.2	164.1	15.3	14.3		10.1	4.8	
Key macroeconomic and fiscal assumptions																	
Real GDP growth (in percent)	0.2	-7.1	-0.6	2.2	3.5	-0.4	4.1	4.2	4.2	4.5	4.8	4.8	4.8	4.5	4.5	4.5	4.5
Average nominal interest rate on forex debt (in percent)	2.2	1.1	1.2	1.1	1.1	1.3	0.5	0.9	0.9	0.8	0.8	0.7	0.7	0.8	1.6	2.5	1.6
Average real interest rate on domestic currency debt (in percent)	6.1	-3.7	3.4	1.6	-6.6	0.2	5.2	-4.1	-1.1	-1.0	-0.9	2.3	1.6	-0.5	-2.0	-2.9	-1.8
Real exchange rate depreciation (in percent, + indicates depreciation)	13.9	-18.4	-12.9	-6.4	10.2	-2.7	14.2	-2.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-5.1	4.8	-2.8	1.7	7.7	1.3	5.2	4.7	1.5	1.3	1.0	1.3	1.7	1.9	2.5	2.5	2.5
Growth of real primary spending (deflated by GDP deflator, in percent)	-10.3	-25.3	17.3	34.4	-19.0	-0.6	25.4	12.7	-6.8	5.6	2.4	-0.5	4.3	2.9	4.0	4.5	4.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	...	...	56.9	56.9	43.0	51.0	51.1	56.9	52.6	56.9	56.9	...

Sources: Country authorities; and staff estimates and projections.

1/ Defined as public external and domestic debt of the central government, inclusive of publicly guaranteed debt of SOEs.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

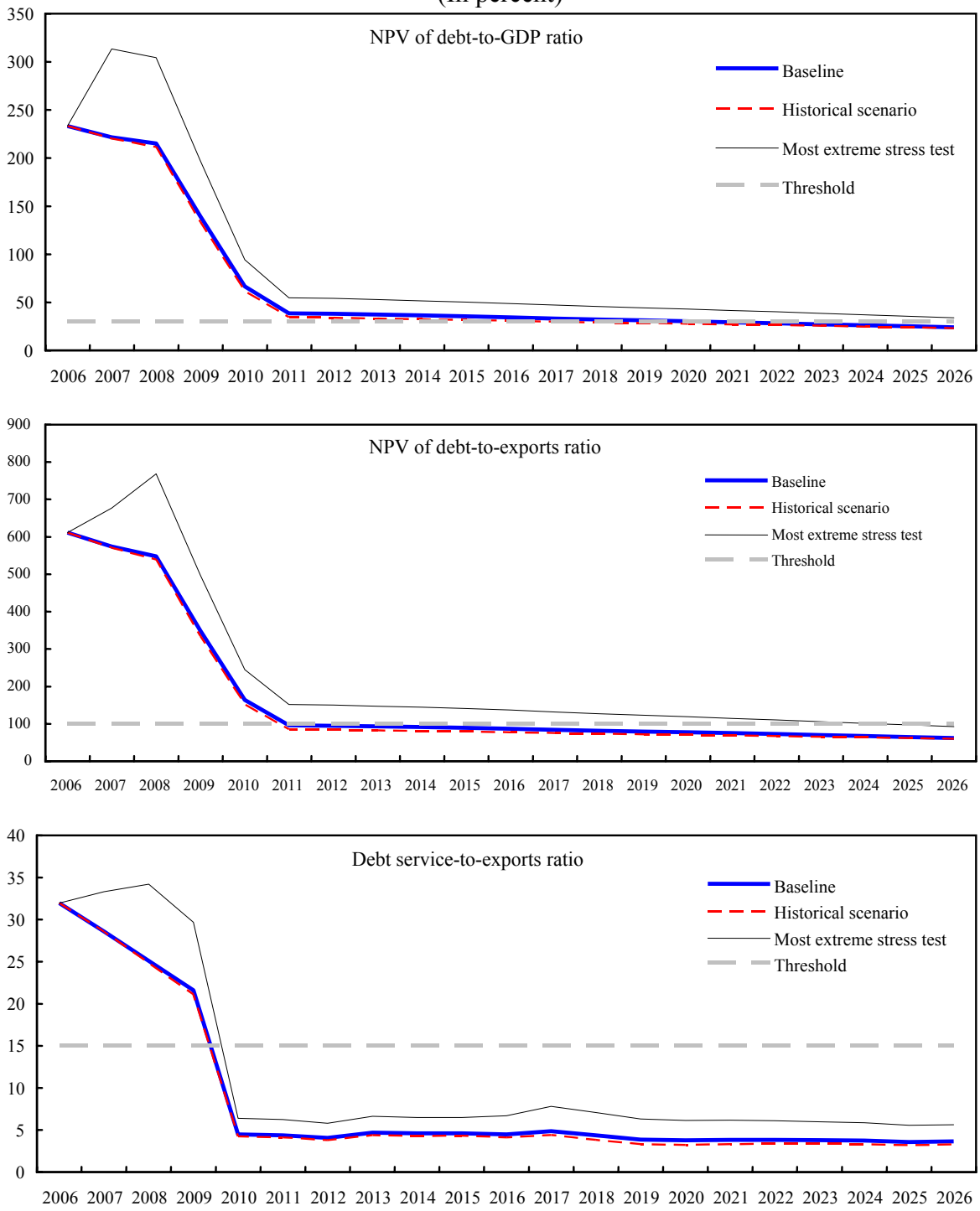
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 3b.Country: Sensitivity Analysis for Key Indicators of Public Debt 2006–26

	Projections							
	2006	2007	2008	2009	2010	2011	2016	2026
NPV of Debt-to-GDP Ratio								
Baseline	282	269	260	179	101	70	57	36
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	282	283	288	210	129	98	112	131
A2. Primary balance is unchanged from 2006	282	270	260	180	103	74	66	53
A3. Permanently lower GDP growth 1/	282	272	265	184	106	75	67	58
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	282	295	313	218	127	92	83	67
B2. Primary balance is at historical average minus one standard deviations in 2007–08	282	272	265	184	106	75	61	38
B3. Combination of B1–B2 using one half standard deviation shocks	282	289	301	207	118	81	63	34
B4. One-time 30 percent real depreciation in 2007	282	370	360	306	195	149	123	79
B5. 10 percent of GDP increase in other debt-creating flows in 2007	282	274	265	183	106	75	61	38
NPV of Debt-to-Revenue Ratio 2/								
Baseline	878	943	901	618	349	244	203	132
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	878	977	967	694	418	314	337	328
A2. Primary balance is unchanged from 2006	878	944	902	621	356	255	234	196
A3. Permanently lower GDP growth 1/	878	949	913	632	361	256	231	202
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	878	1004	1024	712	414	300	279	235
B2. Primary balance is at historical average minus one standard deviations in 2007–08	878	951	917	634	365	259	216	140
B3. Combination of B1–B2 using one half standard deviation shocks	878	992	998	686	389	271	215	119
B4. One-time 30 percent real depreciation in 2007	878	1297	1246	1058	671	517	438	293
B5. 10 percent of GDP increase in other debt-creating flows in 2007	878	959	916	633	364	258	216	140
Debt Service-to-Revenue Ratio 2/								
Baseline	39	41	36	31	10	10	7	5
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	39	42	38	34	12	12	10	13
A2. Primary balance is unchanged from 2006	39	41	36	31	11	10	7	7
A3. Permanently lower GDP growth 1/	39	41	36	32	11	10	8	7
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2007–08	39	43	40	35	12	11	9	9
B2. Primary balance is at historical average minus one standard deviations in 2007–08	39	41	36	31	11	10	7	6
B3. Combination of B1–B2 using one half standard deviation shocks	39	42	39	34	12	11	8	5
B4. One-time 30 percent real depreciation in 2007	39	43	39	34	12	11	8	6
B5. 10 percent of GDP increase in other debt-creating flows in 2007	39	41	36	31	11	10	8	6

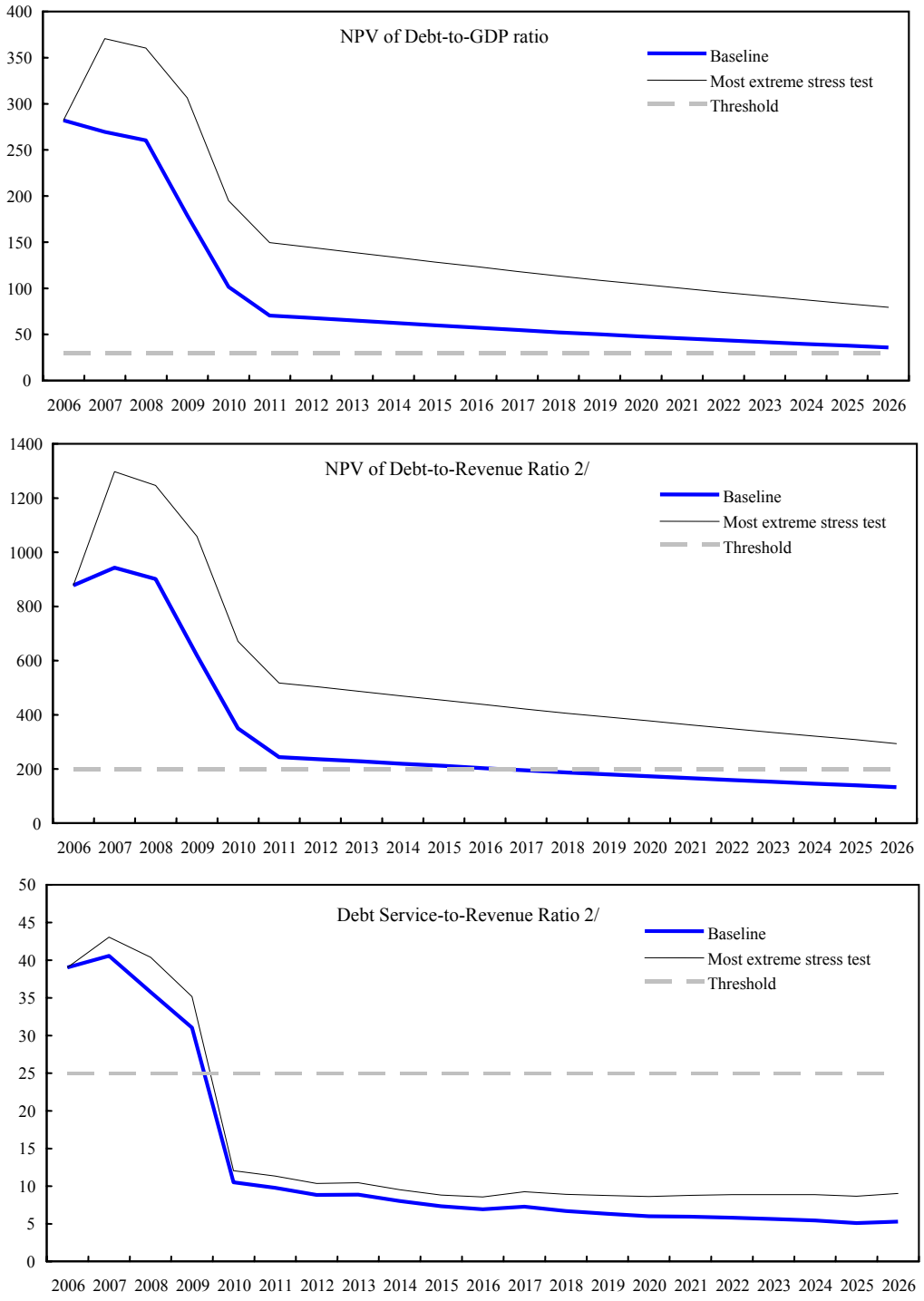
Sources: Guinea-Bissau authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.

Figure 1. Guinea-Bissau: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2006–26
(In percent)



Source: IMF staff projections and simulations.

Figure 2. Guinea-Bissau: Indicators of Public Debt Under Alternative Scenarios, 2006–26 1/



Source: IMF staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2016.
2/ Revenue including grants.